

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Supplementary Winter Examination – 2023**

Branch: Electronics and Telecommunication Engineering/ Electronics and Communication Engineering

Course: B. Tech

Sem.:- VI

Subject with Subject Code: - Digital communication BTETC602

Marks: 60

Date:- 18/01/2024 (2 to 5 PM)

Time:3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain Quantization Noise in PCM.	L1	6
B) What is Delta modulation? Explain its advantages and disadvantages. Explain various noises occurring in Delta Modulation.	L2	6
C) Explain basic Digital communication system with neat block diagram.	L1	6
Q.2 Solve Any Two of the following.		12
A) Differentiate between Pulse Code Modulation (PCM) and Differential Pulse Code Modulation (DPCM)	L3	6
B) Write a note on Wide-sense stationary process.	L2	6
C) Write a note on Additive scramblers. Explain its applications.	L1	6
Q. 3 Solve Any Two of the following.		12
A) What are the cause and effects of Inter Symbol Interference? Explain Eye diagram.	L3	6
B) Explain the working of optimum receiver system with neat block diagram. Also obtain equation for probability of error for the same.	L2	6
C) Explain prominent features of M-ary PSK.	L2	6

Q.4 Solve Any Two of the following. 12

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| A) What is the frequency range, spectral characteristics and application of narrowband noise? | L1 | 6 |
| B) What is Matched Filter? Explain its working with the help of neat block diagram. | L1 | 6 |
| C) Explain Transmission Rates used in T-Carrier system. | L2 | 6 |

Q. 5 Solve Any Two of the following. 12

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| A) Explain the concept of Jammers. What are its applications? | L1 | 6 |
| B) Explain the working of Explain the working of Correlation receiver. | L2 | 6 |
| C) Explain DS-SS technique. What are its advantages and applications? | L2 | 6 |

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